

Variables

A variable is a named container that stores a value in memory. which can be used & changed by the program & it has a fixed type.

How to use: `dataType variableName = Value;`

ex: `int age = 20;`

when to use: whenever you need to store data (input, result, values) to use or change later.

where to use: inside methods, loops, conditions, classes
- anywhere in a program.

why declare a type:

- Java is statically type.
- catches errors early, allocates memory efficiently.
- keeps large code bases safe.

Real-life example: Like labeled jar, the label (name) stays the same, contents (value) can change.

Naming rules:

- case sensitive
- must start with (letter, \$, -) not a digit.
- convention: camel case (ex: studentName).

Date: / /

Type of variables:

TYPE	Declared	Scope
local	Inside a method	only within that method
Instance	Inside class, outside method	Belong to each object
static	Inside class with static keyword	shared across all objects.